

Documentation

Mobile Rig Package (MRP)

TPNVZ06R1300-1A0

Application Group 5B

MS60011/00E



Power. Passion. Partnership.

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1 Safety

1.1 General Service Safety

Contact Authorized Service for Additional Help

If you have any questions about a procedure or safety message related to servicing the engine or the package, contact an authorized service outlet for assistance. Repairs on the system and engine overhaul must be carried out in an MTU authorized workshop.

Trained and Authorized Personnel only

Maintenance and repair work is to be carried out by trained and authorized personnel only. Do not allow unauthorized personnel on or near the system when the system is serviced. Work on or around this system must be performed by people who have the necessary training, skills and tools to do the work. Improper maintenance or repair work can cause severe injury or death for yourself or bystanders.

Using this Manual

Read and understand this safety section before working on this product. This section contains general safety messages. Additional safety messages are included in the specific task procedures.

Signal Words Used in this Manual

Failure to follow the signal words and safety alert symbols contained in this manual can result in serious injury or death. Important safety information is distinguished in this manual by the following notations:

DANGER 	In the event of immediate danger. Consequences: Death or serious injury • Remedial action
WARNING 	In the event of potentially dangerous situations. Consequences: Death or serious injury • Remedial action
CAUTION 	In the event of hazardous situations. Consequences: Minor or moderate injuries • Remedial action
NOTICE 	In the event of a situation involving potentially adverse effects on the product. Consequences: Material damage! • Remedial action. • Additional product information.

Consider the Safety of Procedures You Choose

The manufacturer cannot anticipate every possible circumstance that might involve a potential hazard. If a tool, procedure, work method or operating technique that is not specifically recommended by the manufacturer is used, you must consider possible hazards and satisfy yourself that it is safe for you and for others. Ensure the product will not be damaged or made unsafe by the transport, installation, commissioning, operation, lubrication, maintenance, repair or storage procedures that you choose.

Use Genuine Parts and do not make Unauthorized Modifications

Unauthorized modifications, third-party add-on devices and replacement parts may compromise the integrity of the equipment, and may present a safety risk. Only genuine MTU and DDC parts are allowed to be used to replace components or assemblies. Failure to heed this warning can lead to product damage, serious injury or death and may cause non-compliance with environmental regulations.

Follow Applicable Standards and Regulations

MTU recommends owners, operators and certified technicians become familiar with applicable national safety standards. To reduce the risk of severe injury or death, follow all codes, standards, regulations and laws pertaining to the products installation and application.

Maintain Product Labels

Safety labels should be legible and not covered. If any safety labels are damaged, missing or illegible, contact an MTU-authorized distributor or dealer to obtain replacement labels.

1.2 Use of MTU Technical Documentation

Access to MTU technical documentation is restricted, due to the highly technical nature of the information. Maintenance and workshop documentation is structured based on Qualification Levels (QLs). Personnel are to be qualified at the appropriate level to carry out specific tasks. The QLs are matched to MTU training courses and a demonstrated proficiency related to various tool kits.

QL 1 - Operational monitoring and maintenance tasks that can be done during breaks in operation, and do not require disassembly of the system, engine or any components.

QL2 - Maintenance procedures requiring the exchange of assemblies.

QL3 - Maintenance procedures requiring overhaul of assemblies.

Do not use MTU Technical Documentation or attempt to perform the activities described in this manual if you have not been properly qualified by an MTU-authorized distributor for this QL level.

The information, specifications and illustrations in this manual are based upon information available at the time it was written. The information, specifications and illustrations can change at any time. These changes can affect the service given to the product. Obtain the complete and most current information before you start any work on the system, engine or any components.

If you did not receive this manual through MTU, stop all work and contact MTU to make sure you have the appropriate training and qualifications to follow the procedures within.

Keep this Manual near Engine for Personnel [QL1]

This publication must be provided to all MTU-certified personnel involved in any work being done on the system, engine or components, and a copy must be located in the vicinity of the engine, accessible at all times.

General Responsibilities of Owners and Operators and MTU Technicians [QL1]

Owner and operator responsibilities

In preparation for service of the system, owners and operators must:

- Keep unauthorized personnel clear of system service areas.
- Provide a clear path for transport of the system.
- Provide a service area for the system.
- Provide proper equipment to transport the system.
- Provide that lifting and handling of heavy objects are performed or supervised by personnel knowledgeable with lifting procedures.
- Provide certified forklift and crane operators.
- Provide proper tools for working on the system that are properly maintained and calibrated.
- Provide necessary add-on heat generating devices, such as a fuel heater.
- Make sure that servicing of batteries is performed or supervised by personnel knowledgeable with batteries and required battery service procedures.

1.3 MTU Engine Safety Section

Preparing to Service System

Shut off equipment

You could be seriously injured or killed when working on running equipment. A release or spill of coolant, oil, fuel or hydraulic fluid while the engine is running can cause a fire or explosion. Do not fill coolant or fuel tanks while the engine is running. To reduce the risk of exposure to compressed air, fluids or moving parts, never carry out maintenance and repair work with the system or engine running and:

- Make sure that nobody is standing in the path of moving parts before barring over or starting the system.
- Use care when working around moving belts and rotating parts.
- Remove safety guards only to complete service procedures that can only be completed with the guard removed and replace safety guards as soon as possible once the service is completed. Do not crank or start the engine with the safety guards removed unless specifically required in the procedure.

The engine may need to be running for some types of diagnostic activities.

Let engine and exhaust cool down

All exhaust components are hot when the engine is running and remain hot for an extended period after the engine is shut off. Other components and fluids are hot when the engine is running and remain hot for an extended period after the engine is shut off. Allow all components and fluids to cool before working on or near the unit, or use proper precautions to reduce the risk of being burned.

System placement [QL3]

When placing the system in an area to be serviced, remember the following to reduce your risk of injury or death:

- Place the system on a firm, flat and stable surface capable of supporting the units weight.
- Before servicing anything raised by hydraulics, make sure that it is mechanically supported by stands, blocks or a mechanical lock. Never rely on a hydraulic circuit alone when working underneath raised equipment.

Safe System Servicing

Following safe work practices can help reduce the risk of serious injury or death to yourself or bystanders when servicing a system.

Use proper emergency shutdown procedures

If you have an emergency shutdown situation, use the hot shutdown procedure. You may damage the engine. Each emergency situation is unique. Follow the restart procedures in the manual.

Use lockout and tagout procedures

To reduce the risk of severe injury or death from electric shock or unintentional startup, disconnect all electrical power sources and batteries. Additionally, lockout and tagout the equipment before removing protective shields for service or maintenance. Do not remove a lockout tag unless proper authorization is obtained.

Provide proper ventilation

Do not operate this system in an area where combustible chemicals or vapors may be present. Combustible vapors near the air intake system could be ingested into the engine, causing the engine to suddenly accelerate and overspeed or explode. This condition could cause an unexpected increase in engine rpm or fire. When changing the engine oil or working on the fuel system, make sure that the service area is properly ventilated.

Avoid carbon monoxide poisoning

All engine exhaust contains carbon monoxide, a deadly gas. Breathing carbon monoxide can cause headaches, dizziness, nausea, confusion and eventually death.

Carbon monoxide is a colorless, odorless, tasteless gas that may be present even if you do not see or smell any engine exhaust. Deadly levels of carbon monoxide can collect rapidly and you can be quickly overcome and unable to save yourself. Also, deadly levels of carbon monoxide can linger for hours or days in enclosed or poorly ventilated areas. If you experience any symptoms of carbon monoxide poisoning, leave the area immediately, get fresh air and seek medical treatment.

To prevent serious injury or death from carbon monoxide:

- Never run system in poorly ventilated or partially enclosed areas such as barns, garages, basements, carports, under dwellings, or in pits.
- Never run system outdoors where engine exhaust can be drawn into a building through openings such as windows and doors.
- Make sure that the exhaust pipework is free of leaks and that the gases are discharged into the atmosphere.

Compressed gases and liquids hazards

Remember that you may be exposed to a variety of systems containing gases or fluids under pressure while working on a system. Equipment does not necessarily have to be running to produce high fluid pressure-residual pressures in stationary circuits can represent a serious safety hazard.

Fluids in both liquid and gaseous states can be dangerous when proper safety precautions are not observed. Pressurized systems may cause fires, flying debris, or burns, and escaping fluid under high pressure, even a pin-hole sized leak, can penetrate body tissue, causing serious injury or death.

Electrical hazards

Hazardous voltages are present within this system. Contact with electrical components can result in electric shock. Disconnect all electrical power, and lockout and tagout the equipment before removing protective shields for service or maintenance.

Battery hazards

Refer to battery manufacturers warnings and instructions when interacting with batteries to reduce the risk of injury or death from electric shock, battery fire or explosion. Remember to:

- Use care when disconnecting battery cables. Always remove the negative side of the battery first.
- Charge batteries in a well-ventilated area and do not smoke in battery charging areas.

Using jumper cables

When starting the system with an auxiliary or external power source, ground leads must be connected last and removed first. Improper jumper cable connections can cause an explosion that can result in injury. Connect the ground lead from the power source to the ground lead of the system or to the ground terminal of the starter to avoid sparks in the vicinity of the battery.

Post-Service, Preparing to put System back in Service and to Start Engine

Complete all procedures when preparing to put a system back into service to reduce the risk of serious injury or death to you and bystanders. Keep in mind the following steps to avoid incomplete procedures:

- Make sure all safety guards are in place and secured.
- Check that all protective devices have been reinstalled and all tools removed from the system and/or engine.
- Connect and secure all cables that may have become loose.
- Remove all loose components from the system and engine vicinity.
- Check battery polarity before connecting the cables to the vehicle battery.
- Ground the engine in accordance with the requirements in applicable national, state or local codes.
- Clear the area of bystanders before starting the system.
- Start the engine from the operator station. Never short across the starting motor terminals or the batteries. This could bypass the engine neutral start system or damage the electrical system.
- Complete a successful commissioning inspection.

1.4 MTU General Shop Safety Section

Personal Protective Equipment

Remember, there are many hazards that you may encounter in the workplace, such as falling objects, flying debris, loud noises, chemical exposure and sharp edges. To reduce your risk of injury when servicing the system and/or engine, always consider your activity and remember to wear appropriate protective gear:

- Feet – approved safety shoes (not bare feet, sandals or sneakers)
- Face – approved face shield
- Eyes – protective glasses or goggles
- Ears – hearing protection
- Lungs – respiratory protection
- Head – safety helmet, tied back hair
- Hand – gloves (for example, sleeved heat protective gloves)
- Body – protective clothing – apron, cotton work clothing (no rings, watches, jewelry or loose clothing)

Use of Compressed Air

Follow general shop safety procedures when utilizing compressed air and follow these common rules to reduce your risk of injury:

- Always use a pressure reducing valve and a safety valve and do not exceed 376 kPa (40 psi) air pressure. Make sure that the connected devices and equipment are appropriate for this pressure.
- Make sure that the hose coupling and connections are securely attached together.
- Make sure that the snout of the air nozzle has a protective disc (for example, rubber disc) to prevent air-borne particles from being deflected.
- Always shut off compressed air lines before compressed air equipment is disconnected from the supply line or before equipment or tools are exchanged.
- Never use compressed air to force flammable liquids out of containers.
- Never force compressed air into thin-walled containers (for example, containers made of tin, plastic or glass) for drying purposes or to check for leaks as the container may shatter.
- Do not direct a compressed-air jet at someone and never use compressed air to clean contaminated clothing while it is worn.

Safe Use of Tools and Equipment

Make sure you understand how to use the tools before performing any service work. Always use tools that are in good conditions and follow manufacturers instructions. Improper use of tools or use of tools in poor condition can increase your risk of injury.

Working from Heights

When working from heights on the system, always use proper procedures for working at height and follow equipment manufacturers instructions. To reduce your risk of injuries from falls while working on MTU systems and engines, remember to:

- Use suitable ladders and work platforms.
- Make sure components are placed on stable surfaces.
- Never use pressurized lines for climbing or support.

Welding

When welding, always use proper welding techniques and follow the manufacturers instructions and warning. To reduce your risk of injuries from welding, also keep in mind:

- If components (for example, exhaust manifold) are to be welded, they must be removed from the engine.
- Do not use the assembly or system as ground terminal.
- It is not necessary to remove the connector and the connections when carrying out welding operations on the manufacturers electronics if the master switch for power supply is switched from "ON" to "OFF" and the wire is disconnected from the negative and positive poles on the battery.

Safe Work Space

As with other industrial work, it is important to maintain an area surrounding the engine that is dry, well lit, ventilated and free from clutter, loose tools, parts, ignition sources and hazardous substances. Be aware of hazardous conditions that can exist. Make sure a First Aid Kit is available.

Fluids and lubricants must be kept in suitable, properly designated containers. Never put maintenance fluids into glass containers – glass containers can break.

Cleanup

After completion of maintenance and repair work, make sure that no loose objects are in or on the assembly or system. Try to organize parts in bins and on benches when you are disassembling components.

To reduce the risk of injury from slipping and falling, clean up spilled liquids with suitable cleaning agents or as defined by the manufacturers specifications. Clean up oil spills quickly. You can do this by applying absorbent grit: this not only absorbs oil but makes it less likely that a person may slip and fall on an oil slick.

Proper Disposal

Always dispose of all lubricants (motor oil, coolant, gear box oils, and so on.) and filters according to federal or local regulations. Used oil disposed of in nature or waterways contaminates drinking water and kills wildlife.

Fire Prevention

Ignition sources

As in other industrial contexts, there are various potential sources of fire or explosion hazards when working on a system or an engine. To reduce the risk of injury from fire or explosion, be aware of potential ignition sources. Some examples of ignition sources include:

- Electrical sparks or arcs
- Cutting, welding or grinding sparks
- Open flame
- Hot surfaces
- Incandescent lights
- Smoking

Flammable materials and vapors

Avoid exposing potential ignition sources to flammable materials and vapors. Some examples of flammable materials and vapors include: